UPS V4 — complete prototype schematic / Phase 3.2 owner review

F1 prototype: Littelfuse 0451005.MRL, 5 A, 125 Vdc; qualify temperature, cable, hot-plug, polarity-FET SOA and coordination.

VIN damping R12 2 ohm / C26 2.2uF 100V: selected prototype parts; VAC1 hot-plug qualification OPEN.

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

PGND = system return; GND = quiet TS / PROG / NTC; NT1 local tie; no PCB moat.

VIN hot-plug RC: TI Fig. 8-1 (>15V)
Provisional R12 / C26; validate on prototype

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

SERVICE USB / ESP32-C3 RECOVERY — prototype, not fabrication approval

U3 SYS -> MAIN_3V3 buck; U4 USB_VBUS -> USB3V3 LDO; U5 normal-priority mux -> MCU3V3.

USB-only ROM flashing: BOOT held low during RESET; no battery/adapter required. Wi-Fi off in service firmware.

Complete power interruption: EN 10k/1u RC and GPIO straps restart MCU. Transition resets acceptable; hardware tests pending.

U1 project footprint corrected to 61 lands / 53 numbers; verify RF keepout, stencil and placement in PCB phase.

Optional independent 5V F5 block is NOT IMPLEMENTED; resolve F0-only versus common-PCB F5 before PCB freeze.

Q5/Q6/Q7 default off; Q3 sinks CE only on request. Q6 is not a precision MAIN voltage monitor.

CE pull-up via REGN/MAIN diode OR. BQ25792 2S charger-only reset/default 2A risk OPEN.

Sheet: /
File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

Size: A1 Date: **Rev: 2.1**
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